Cover Letter

Jonathan Herrera-Vásquez
Universidad Americana de Europa (UNADE), Department of Computer Science
Cancún, Quintana Roo, Mexico

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Dear Editor,

I wish to submit an original research article entitled “When Does SHAP Outperform LIME? Model- and Configuration-Dependent Results from a Paired Tabular Benchmark” for consideration by Computers and Informatics.

I confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere. An earlier version was submitted to another journal and has been withdrawn from it before this submission.

In this paper, I report a paired comparison of LIME and SHAP, the two explanation methods most used for tabular machine-learning models, in which both methods explain the same model, the same instances and the same random seed. This pairing separates a difference between the methods from a difference between the models or samples they were run on, which is rarely done. The comparison shows that SHAP is more stable and more faithful to the model, but also that two common rules of thumb need a condition: the latency ordering of the two methods depends on the model family and reverses between two tree ensembles, and the low stability of LIME on one dataset depends on its kernel width and on the feature space. This is significant because practitioners choose between the two methods on such rules, and a ranking obtained on one dataset and one configuration does not transfer by default.

I believe that this manuscript is appropriate for Computers and Informatics because it contributes to artificial intelligence, one of the computer-science areas of the journal’s Aims & Scope, with an empirical, reproducible study of explainable machine learning that ends in a deployment recommendation conditioned on the model architecture. The code and all run artifacts are public.

Relation to earlier work. The runs analysed here were released with an earlier article by the author and a co-author (Revista de Investigación Multidisciplinaria Iberoamericana, https://doi.org/10.69850/rimi.vi3.307), which includes a paired SHAP–LIME test on a 45-cell subset of three model families. The present manuscript extends that test to the full 75 cells, adds confidence intervals and effect sizes, and contributes the conditional results above, which are not in that article; it restates none of its numerical results. Other manuscripts by the author, under review at other journals, analyse the same runs for different questions and report none of the results reported here. The code, the run artifacts and earlier drafts of this work are in a public repository, archived at https://doi.org/10.5281/zenodo.23147228; this may show in the similarity report.

I have no conflicts of interest to disclose.

Please address all correspondence concerning this manuscript to me at jonnabio@gmail.com.

Thank you for your consideration of this manuscript.

Sincerely,
Jonathan Herrera-Vásquez